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Motorized Haptic Resistance Controller for Microtonal Performance Interfaces
(Series D)

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Inventors: Dr. Soren Lindqvist • Assignee: Zazie Institute of Applied Anomalies / Zazie Productions LLC

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ABSTRACT & METHOD DISCLOSURE

A musical performance controller comprising motorized linear faders and touch keys programmed with closed-loop force-feedback, physically simulating variable mechanical spring tension and scale detents.

INDEPENDENT CLAIMS

1. An apparatus for experimental sound transduction, comprising: an acoustic input transducer configured to convert physical vibrations into an analog electrical signal; a dynamic hysteresis conditioning stage; and an acoustic output transducer configured to project conditioned sound into an architectural boundary.

2. The apparatus of claim 1, further comprising a real-time digital microcontroller configured to monitor spectral peaks and adjust closed-loop phase alignment.

3. A method for algorithmic spatial sound diffusion, comprising: sampling multichannel audio streams; calculating distance-based amplitude weights across distributed speaker coordinates; and routing low-latency audio packets over an asynchronous IP network.

DEPENDENT CLAIMS

4. The apparatus of claim 1, wherein the dynamic conditioning stage comprises a soft-clipping JFET transistor network.

5. The apparatus of claim 1, wherein the acoustic output transducer comprises an exponential horn with aluminum phase plugs.

6. The method of claim 3, wherein the network protocol is Dante operating under IEEE 1588 precision time protocol.

7. The apparatus of claim 1, wherein the input transducer comprises a dual-axis optical laser triangulation sensor.

PRIOR-ART CRITIQUE

Contrasted with traditional commercial DAW plugins and static acoustic room treatments, the present speculative invention couples physical hardware dynamics directly to room boundaries, enabling adaptive generative performance without artificial digital latency.

COUNSEL MEMO

Defensive publication strategy approved under ZIAA Open Research Charter. Discloses all core mathematical and physical topologies to prevent predatory patent encumbrance by proprietary audio conglomerates.